Technical Specification of Final Project of Youth Skill Development Training Program Nov'24 ~ Feb'25 AI Team

Project Title: Conversational Memory Bot – AI-Powered Photo Gallery Assistant

Project Scope:

The Conversational Memory Bot is an AI-powered chatbot, designed to revolutionize the way users interact with their personal photo galleries. By combining advanced Natural Language Processing (NLP) and other frameworks, this system enables users to query, retrieve, and explore their photos using natural language and visual features.

Core Functionalities:

1. Natural Language Querying: Users can ask questions like, "Show me photos of dog playing in the park" or "Find pictures from a trip to Italy in 2021."
2. Contextual Image Retrieval: The bot retrieves relevant photos based on contextual understanding of the query, matching both text and image features.
3. Detailed Image Descriptions: Generate descriptions of selected images, such as "This image shows a beach during sunset with two people walking."
4. Visual Similarity Search: Find images visually similar to a selected one based on color, object, and scene similarity.
5. Automatic Tagging and Organization: Automatically tag photos with keywords like "beach," "sunset," or "dog," for easier searching and organization. [Optional]

Scope of Image Types:

- **Travel Memories:** Retrieve and discuss photos from specific trips or locations.
- **Event Cataloging:** Group and query photos from birthdays, weddings, concerts, or specific events.
- **Object and Activity Search:** Search for objects (e.g., cars, books) or activities (e.g., hiking, swimming) in images.
- **Pet Photos:** Identify and discuss photos of pets by type or specific pet names.
- **Nature and Landscapes:** Retrieve photos of sunsets, forests, beaches, and other natural scenes.
- **Style and Color:** Search for images based on predominant colors or visual styles.

Features Details:

1. Natural Language Query

- Description: Users can interact with the system using everyday language to ask questions about their images.

2. Contextual Image Retrieval

- Description: Match user queries with images based on contextual and semantic understanding of the query and metadata.

3. Image Descriptions

- Description: Generate detailed descriptions of selected images, including objects, activities, and settings.

4. Visual Similarity Search

- Description: Retrieve images visually similar to a selected one by comparing features like color, texture, and objects.

5. Automatic Tagging [Optional]

- Description: Use AI to automatically tag images with relevant keywords based on detected objects, scenes, activities, and people.

6. Relevance Ranking

- Description: Rank retrieved images based on how closely they match the user’s query in context and meaning.
- Implementation: Use a scoring mechanism combining NLP and image features for ranking.

7. Interactive Gallery

- Description: A gallery interface allowing users to browse images with advanced interactivity:
- Image Description Modification: If user provides or modify more information about an image during conversation the AI will detect and modify the description based on the user’s proposition. [Optional]

Technology and Components

1. Natural Language Processing (NLP)

- **Use Case:** Interpreting user queries and mapping them to image content.
- **Tools:** Vision based LLM/SLM
- **Functionality:**
    - Tokenization and semantic understanding of text queries.
    - Query expansion to handle synonyms or rephrased requests.

3. Embeddings

- **Use Case:** Representing text and image features in a shared vector space for matching and similarity.
- **Tools:** Sentence Transformers
- **Functionality:**
    - Text embeddings for NLP queries.
    - Image embedding for visual feature representation.

4. Retrieval-Augmented Generation (RAG)

- **Use Case:** Enhance query results with generative AI by combining relevant image data with natural language responses.
- **Tools:** Any RAG frameworks
- **Functionality:**
    - Retrieve matching images using embeddings.
    - Generate human-like responses about the retrieved images.

5. Large Language Models (LLM)

- **Use Case:** Generate contextual and natural language responses to user queries.
- **Tools:** llava (https://ollama.com/library/llava)

- **Functionality:**
    - Summarize or elaborate on image content.
    - Provide explanations and recommendations.

6. Information Retrieval

- **Use Case:** Efficiently retrieve relevant images from large datasets.
- **Tools:** TBD
- **Functionality:**
    - Fast indexing and querying of image metadata and embeddings.

7. Image Feature Extraction

- **Use Case:** Extract meaningful features from images for tagging and similarity matching.
- **Tools:** Prompt engineering on vision based LLM/SLM
- **Functionality:**
    - Generate embedding of the features

8. Semantic Understanding

- **Use Case:** Combine text and image data to ensure accurate query interpretation.
- **Tools:** Multimodal models
- **Functionality:**
    - Align textual and visual information for relevance.

9. Multimodal AI (Multimodal RAG)

- **Use Case:** Leverage both text and visual inputs to enhance user interaction.
- **Tools:** A Multimodal Models.
- **Functionality:**
    - Combine user text queries and images to improve context.

System Workflow

1. Query Processing

1. User enters a query (e.g., "Show me photos of cat in the garden").
2. NLP module interprets the query and generates embedding.

2. Image Retrieval

1. Extract image embeddings for gallery images.
2. Match text embedding with image embedding using similarity scores.

3. RAG Pipeline

1. Retrieve top-ranked images using semantic search.
2. Generate contextual responses based on user queries and retrieved images.

4. Output

1. Display results in an interactive gallery with options for zoom, tags, and additional details.

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Deliverables:

1. SRS, WBS
2. WEB UI (Pages: Homepage, Gallery, Batch Image Uploader, Image viewer, Chat)
3. PPT
4. Codebase (GitHub Repo)

Marking Criteria:

Requirements Understanding - 10

Functionality Completeness - 20

Expected Output/Efficiency - 30

Code Quality - 20

Demonstration - 10

Documentation (SRS, WBS) - 10